

NLP Fundamentals: Task & Challenge Solutions

Task 3: Build BoW and TF-IDF from Scratch

Input Documents

```
docs = [  
    "I love cricket",  
    "I love football",  
    "Cricket is my favorite sport"  
]
```

Part A: Preprocess

We convert the text to lowercase, remove punctuation, and drop common stop words (like "I", "is", "my") to focus on the core meaning.

```
[  
    ['love', 'cricket'],  
    ['love', 'football'],  
    ['cricket', 'favorite', 'sport']  
]
```

Part B: Create Vocabulary

```
[  
    'love',  
    'cricket',  
    'football',  
    'favorite',  
    'sport'  
]
```

Part C: Build Bag of Words Table

Document	love	cricket	football	favorite	sport
Doc1	1	1	0	0	0

Document	love	cricket	football	favorite	sport
Doc2	1	0	1	0	0
Doc3	0	1	0	1	1

Part D: TF-IDF Analysis

1. Which word gets the highest TF-IDF score?

The words **"football"**, **"favorite"**, and **"sport"** will tie for the highest TF-IDF scores within their respective documents.

2. Why?

TF-IDF rewards words that appear frequently in a specific document but rarely across the entire corpus. These three words each appear in exactly one document (high Inverse Document Frequency) and constitute a significant portion of their respective documents (high Term Frequency).

3. Why does "love" get a lower score?

Because "love" appears in multiple documents (Doc1 and Doc2). Its Document Frequency is higher, meaning its Inverse Document Frequency (IDF) weight drops, penalizing it for being a more common word across the corpus.

4. Which words appear in only one document?

"football" (only in Doc2), **"favorite"** (only in Doc3), and **"sport"** (only in Doc3).

🎯 Learning Outcome Achieved:

- ✓ Vocabulary Creation
- ✓ Bag of Words representation
- ✓ Understanding Term Frequency - Inverse Document Frequency (TF-IDF)
- ✓ Importance of Rare Words in extracting semantic value

Mini Challenge (Interview Style)

Given Text: **"Oh great, another Monday. LOL."**

1. What preprocessing steps will be applied?

Standard steps include: Lowercasing, removal of punctuation (commas, periods), tokenization, and potentially removing stopwords (like "another"). Advanced steps might include slang normalization (mapping "LOL" to "laugh" or "funny").

2. What tokens remain after cleaning?

Depending on the strictness of the stopwords list, the remaining tokens are likely: ['oh', 'great', 'monday', 'lol']. If "oh" is treated as a stopwords, it would just be ['great', 'monday', 'lol'].

3. Which NLP challenge is present?

The primary challenges here are **Sarcasm / Irony** and handling **Internet Slang** ("LOL").

4. Why might a sentiment model classify it incorrectly?

A naive sentiment model looks at the polarity of individual words. The words "great" and "LOL" carry highly positive weights. The model will likely aggregate these and output a "Positive" sentiment, entirely missing the context of annoyance conveyed by "another Monday."

5. Would BoW understand the sarcasm? Why or why not?

No. Bag of Words entirely ignores word order, context, and semantic relationships. It treats the sentence as an unordered bucket of terms. Without sequence and context, the contrast between the literal positivity of "great" and the situational negativity of "another Monday" is completely lost.